

Suggested Teaching Guidelines for

Data Analytics
PGCP-AI August 2026

Duration: 120 Hours: 50 Theory (T) + 50 Lab (L) + 20 Self-Learning (SL)

Objective: To introduce the student to Data Analytics

Prerequisites: Knowledge of Python Programming and basics of statistics.

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment– 20%

List of Course Material

Text book: Business Analytics James R Evans, 2nd Edition, Pearson

Reference Book:

1. Business Analytics: Data Analysis & Decision-Making Hardcover – Widescreen
2. Data Science for Business, Foster Provost & Tom Fawcett
3. Data Science and Analytics Paperback by V. K. Jain (Author)
4. Data Analytics for Beginners: A Beginner's Guide to Learn and Master Data Analytics, by Robert J. Woz
5. Data Analytics: Become A Master in Data Analytics by Richard Dorsey

Note:

- ***Lab assignments are indicative; faculty shall provide adequate assignments for better practice.***

Sessions 1 & 2: (4T + 4L + 2SL)

Theory:

- Introduction to Business Analytics using some case studies
- Data analytics Life Cycle: Discovery, Data preparation, Model planning
- Model building implementation
- Quality assurance, Documentation, Management approval
- Installation
- Acceptance and operation

Lab Assignment: Setup workspace in either in Google Colab or Jupyter and go through the documentation of Pandas, Numpy, Sklearn libraries.

Session 3: (2T + 2L + 2SL)

Theory:

- Intelligent data analysis and Nature of Data
- Analytic Processes and Tools
- Analysis vs. Reporting
- Modern Data Analytic Tools

Lab Assignment: Explore and analyze the nature of data.

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Session 4: (2T + 2L + 2SL)

Theory:

- Visualization and Exploring Data
- Case studies: Making Right Business Decisions based on data

Lab Assignment: Load a sample dataset and explore the data and draw some insights. Also use Matplotlib and seaborn libraries.

Session 5: (2T + 2L + 2SL)

Theory:

- Descriptive Statistical Measures
- Summary Statistics - Central Tendency & Dispersion (Mean, Median, Mode, Quartiles, Percentiles, Range, Interquartile Range, Standard Deviation, Variance, and Coefficient of Variation)

Lab Assignment: Load any dataset and find out the mean, median mode and other central tendencies of the dataset.

Sessions 6 & 7: (4T + 4L + 2SL)

Theory:

- Sample & population, Uni-variate and bi-variate sampling, re-sampling
- Sample Spaces and Events
- Joint, Conditional and Marginal Probability
- Bayes' Theorem

Lab Assignment: Load any dataset and apply Naive Bayes function and predict the output.

Session 8: (2T + 2L + 2SL)

Theory:

- Random Variable
- Probability Distribution and Data
- Continuous and discrete distribution – (Normal, Binomial, and Poisson distribution)
- Central Limit Theorem

Lab Assignment: Generate random numbers and check if they are in normal distribution using scipy libraries.

Session 9: (2T + 2L)

Theory:

- Sampling and Estimation
- Statistical Interfaces

Lab Assignment: Load any dataset and explore sampling techniques

Session 10: (2T + 2L)

Data Analytics**PGCP-AI August 2026****Theory:**

- Concepts of Correlation
- Covariance
- Pearson Correlation
- Outliers

Lab Assignment: Load any dataset and find out the covariance between two fields and also find the correlation and determine how two fields are correlated. Also handle the outliers in the data.

Sessions 11 & 12: (4T + 4L +2SL)

- Hypothesis Testing
- Z-Test
- Chi-Square test
- Skewness

Session 13: (2T + 2L)

- Predictive modelling and analysis
- Application
- Types
- Benefits and challenges
- The Future of predictive modelling
- The Limitations of Predictive modelling
- Predictive modelling Tools

Sessions 14, 15 & 16: (6T + 6L)

- Predictive Modelling (From Correlation to Supervised Segmentation):
- Identifying Informative Attributes
- Segmenting Data by Progressive Attributive
- Models Induction and Prediction
- Supervised Segmentation
- Visualizing Segmentations
- Trees as Set of Rules
- Probability Estimation, Prescriptive Modelling
- Difference between predictive and prescriptive modelling
- Working of prescriptive analytics
- Examples and use cases

Sessions 17 & 18: (4T + 4L)

- Regression Analysis
- Forecasting Techniques

Sessions 19 & 20: (4T + 4L)

- Simulation and Risk Analysis
- Optimization- Linear, Nonlinear

Session 21: (2T + 2L)

- Overfitting and Its Avoidance:
- Generalization
- Holdout Evaluation Vs. Cross Validation

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Session 22: (2T + 2L)

- Decision Analytics
- Evaluating Classifiers
- Analytical Framework
- Evaluation
- Baseline
- Performance and Implications for Investments in Data

Session 23: (2T + 2L)

- Evidence and Probabilities
- Explicit Evidence Combination with Bayes Rule
- Probabilistic Reasoning

Session 24: (2T + 2L)

- Factor Analysis
- Directional Data Analytics
- Functional Data Analysis

Session 25: (2T + 2L)

- Introduction to Power BI architecture and interface
- Importing and transforming data using Power Query
- Creating data visualizations using charts and reports
- Designing interactive dashboards and sharing insights